University *of* Massachusetts Amherst
Department of Physics

Manuel Díaz

Curriculum Vitae

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EDUCATION

Ph.D. Candidate in Physics, University of Massachusetts Amherst — 2020 – Present

Amherst Center for Fundamental Interactions (ACFI)
Advisor: David Kastor. Expected completion: May 2027.
Dissertation working title: Einstein Gravity and Beyond: From Black Hole Thermodynamics to Inflationary and Quantum Cosmology

M.S. in Physics, Pennsylvania State University — 2017 – 2019

Thesis: Semiclassical Consistent Constraints with Moments in Spherically Symmetric Quantum Gravity
Advisor: Martin Bojowald

B.S. in Physics; B.S. in Mathematics, University of Texas at Austin — 2012 – 2017

RESEARCH EXPERIENCE

Doctoral Research, University of Massachusetts Amherst — Present

My doctoral research studies general relativity and modified gravity with second order field equations. With David Kastor, I study black hole thermodynamics when charged membranes separate regions with different gravitational couplings, and the membrane ergospheres that result from them. With Martin Bojowald, I study canonical gravity and its application to quantum cosmology. With Jennie Traschen, I study the effects of black holes on the duration of inflation.

Master's Research, Pennsylvania State University — 2017 – 2019

With Martin Bojowald, I studied quantum gravity corrections to general relativity using canonical effective methods. I showed that semiclassical constraint equations for spherically symmetric spacetimes respect general covariance in the Hamiltonian formalism.

Undergraduate Research, University of Texas at Austin — 2012 – 2017

With J. Craig Wheeler, I modeled massive star evolution with MESA to study Betelgeuse's rapid rotation, which suggested that Betelgeuse may have partially or completely swallowed a companion star.

PUBLICATIONS

M. Bojowald, M. Díaz, and E. I. Duque, *"Perturbative emergent modified gravity on cosmological backgrounds: Kinematics."* Physical Review D 113, 084061 (2026). arXiv:2507.14358.

M. Bojowald, M. Díaz, and E. I. Duque, *"Singularities in loop quantum cosmology."* Physical Review D 113, 084060 (2026). arXiv:2507.08116.

K. Berglund, M. Bojowald, A. Colter, and M. Díaz, *"Spacetime superpositions as fluctuating geometries."* Physical Review D 111, 044021 (2025). arXiv:2405.20193.

K. Berglund, M. Bojowald, M. Díaz, and G. Sims, *"Quasiclassical solutions for static quantum black holes."* Physical Review D 109, 024006 (2024). arXiv:2012.07649.

S. Nance, J. M. Sullivan, M. Díaz, J. C. Wheeler, *"The Betelgeuse Project II: Asteroseismology."* Monthly Notices of the Royal Astronomical Society 479, 251–261 (2018). arXiv:1805.10347.

J. C. Wheeler, S. Nance, M. Díaz, S. G. Smith, J. Hickey, L. Zhou, M. Koutoulaki, J. M. Sullivan, J. M. Fowler, *"The Betelgeuse Project: Constraints from Rotation."* Monthly Notices of the Royal Astronomical Society 465, 2654–2661 (2017). arXiv:1611.08031.

Manuscripts in Preparation

M. Díaz and D. Kastor, "Black Hole Enthalpy and Membrane Ergospheres with a Varying Einstein Coupling." Manuscript in preparation.

M. Díaz, D. Herrera, D. Kastor, and B. Schmidt, "Membrane Ergospheres and Mining Energy from Born–Infeld Black Holes." Manuscript in preparation.

S. Corbà, M. Díaz, and J. Traschen, "Effects of Black Holes on Inflation." Manuscript in preparation.

M. Bojowald, M. Díaz, and E. I. Duque. "Perturbative emergent modified gravity on cosmological backgrounds: Tensor-Vector-Scalar modes."

M. Bojowald, M. Díaz, and E. I. Duque. "Strictly non-singular perturbative quantum cosmology."

M. Bojowald, M. Díaz, E. I. Duque, and S. Murk. "Scalar power spectrum in bouncing cosmologies."

M. Bojowald, M. Díaz, E. I. Duque, and Ö. Güleryüz. "Emergent modified gravity: Dynamical dark energy without additional degrees of freedom."

PRESENTATIONS

"Emergent modified gravity: Dynamical dark energy model without additional degrees of freedom" May 2026
Phenomenology 2026 Symposium (PHENO), University of Pittsburgh

"Perturbative Emergent Modified Gravity in Cosmological Backgrounds" Oct 2025
COSMO-25, Carnegie Mellon University

"Perturbative Emergent Modified Gravity in Cosmological Backgrounds" Jul 2025
Quantum Gravity 2025, Penn State

"Gauge-Invariant Observables for Gravitational Waves from a First-Order Electroweak Phase Transition" May 2025
Phenomenology 2025 Symposium (PHENO), University of Pittsburgh

"Quasiclassical Solutions for Static Quantum Black Holes" May 2024
APS Division of Particles & Fields (DPF), University of Pittsburgh

Guest Lectures and Panels

"Introduction to Black Holes", College of the Holy Cross Mar 2026
Invited guest lecture for undergraduate astrophysics course (Prof. K. Berglund), Worcester, MA

Guest, Autism Stories Podcast, Autism Personal Coach Nov 2021
Discussed physics, neurodiversity, and resources in Spanish for the autistic community

"Gravitational Waves", PaTTAN Autism Conference, Penn State Aug 2018
Guest lecture on gravitational waves for neurodivergent high school students, Penn Stater Conference Center

Conference Panelist, Autism and Postsecondary Transition, PaTTAN, Penn State Aug 2018
Panel on postsecondary transition for neurodivergent students; audience of educators, families, and students.

"Autism and Neurodiversity", Molotov Seminar, UT Austin Mar 2016
Public talk on autism and the neurodiversity movement, hosted by Texas Neurodiversity

Poster Presentations

"The Betelgeuse Project: Constraints from Rotation" Jan 2017
229th Meeting of the American Astronomical Society (AAS), Grapevine, TX

TEACHING EXPERIENCE

Teaching Assistant, UMass Amherst 2021 – 2026
Introductory physics sequences and laboratories for life science, engineering, and physics majors. Courses also include general relativity, quantum mechanics, and computational physics.

SCHOOLS, WORKSHOPS, AND CONFERENCES

ACFI Workshop: Black Holes and Horizons in Quantum Gravity, UMass Amherst — Oct 2026
Phenomenology 2026 Symposium (PHENO), University of Pittsburgh — May 2026
COSMO-25, Carnegie Mellon University — Oct 2025
ACFI Workshop: Primordial Black Holes, UMass Amherst — Sep 2025
Quantum Gravity 2025, Penn State — Jul 2025
Phenomenology 2025 Symposium (PHENO), University of Pittsburgh — May 2025
TASI 2024 – The Frontiers of Particle Theory, University of Colorado Boulder — Jun 2024
APS Division of Particles & Fields (DPF), University of Pittsburgh — May 2024
Surveying the Landscape, UMass Amherst — Apr 2024
SACNAS National Diversity in STEM (NDiSTEM) Conference — Oct 2023
Bard Summer School on Quantum Gravity, Bard College — Jun 2019
PaTTAN Pennsylvania Autism Conference, Penn Stater Conference Center — Aug 2018
229th Meeting of the American Astronomical Society, Grapevine, TX — Jan 2017
Autism Campus Inclusion (ACI) Leadership Academy, ASAN, Washington, D.C. — Jun 2016

EQUITY, DIVERSITY, INCLUSION, AND SERVICE

Research Mentor, UMass Amherst — 2026 – Present
Mentor a master's student and an undergraduate in David Kastor's group on research and professional development.

Graduate Representative, Physics Climate Committee, UMass Amherst — 2023 – 2025
Assessed and improved departmental climate; advocated for inclusive policies.

Member, SEA Change Working Team (AAAS), UMass Amherst Physics — Spring 2025
Contributed to a departmental review and developed an action plan spanning five years to improve climate, equity, and inclusion as part of the AAAS STEM Equity Achievement Change initiative.

Graduate Student Mentor, UMass Amherst — 2021 – Present
Guide incoming graduate students in their first year on coursework, advisor selection, and professional development.

Volunteer Mentor Judge, SACNAS NDiSTEM Conference — 2023
Professional development and networking focused on increasing representation in STEM.

Bilingual STEM Outreach Volunteer, AstroFest, Penn State — 2017 – 2019
Delivered bilingual (English/Spanish) interactive astronomy activities to children and families.

STEM Outreach Volunteer, Penn State summer science camps — 2017 – 2018
Delivered interactive physics lessons for elementary school students, including a lecture on gravitational waves adapted for young learners.

Founder and President, Texas Neurodiversity, UT Austin — 2015 – 2017
Founded the first neurodiversity advocacy organization at UT Austin. Organized Sensory Friendly Concerts and educational talks on Autism Acceptance. Selected for the 2016 Autism Campus Inclusion Leadership Academy.

Neurodiversity Writer and Translator — 2015 – 2017
Founded Neurodivergencia Latina and helped found Autismo, Liberación y Orgullo. Translated neurodiversity advocacy works into Spanish.

AWARDS AND HONORS

Selected Participant, CNS EDGE Program, UMass Amherst 2026 – 2027
Enhancing Diversity in Graduate Education, College of Natural Sciences.

CNS Summer Research Assistantship, UMass Amherst Summer 2026

CNS Diversity Travel Grant, UMass Amherst 2023
Awarded by the College of Natural Sciences Dean's Office to attend the SACNAS NDiSTEM Conference.

Bunton–Waller Fellowship, Penn State 2017 – 2019
Graduate research fellowship awarded to students who enhance diversity.

Autistic Self Advocacy Network – Inaugural Autistic Scholars Fellowship 2016
National tuition scholarship recognizing commitment to disability rights advocacy and Autistic culture.

CNS ASPIRE Achievement Award for Leadership and Service 2016
Awarded by the Black and Latinx Advocacy Council, UT Austin, for leadership and service.

Nellie H. and Oscar L. Roberts Scholarship Fund, Penn State 2018 – 2019

Homer F. Braddock Scholarship in Biology, Chemistry and Physics, Penn State 2017 – 2018

Walter E. Millett Endowed Undergraduate Scholarship in Physics 2016

Kevin E. Underhill Memorial Endowed Presidential Scholarship 2015 – 2016

Retired Faculty and Staff Association (RFSA) Scholarship 2015 – 2016

Melvin J. Rieger Undergraduate Scholarship in Physics 2014 – 2015

The Will Rogers Memorial Scholarship 2013 – 2016

College of Natural Sciences ASPIRE Professor's Choice Award 2013
Presented in recognition of academic curiosity, perseverance, and achievement at the University of Texas at Austin.

Freshman Research Initiative Fellowship 2013

President's Achievement Scholarship 2012 – 2016

Robert E. Boyer Endowed Presidential Scholarship 2012 – 2015

SKILLS

Languages: English, Spanish (native bilingual)

Programming and Tools: Wolfram Mathematica, Python, Fortran, C/C++, LaTeX, MATLAB, R, FormCalc, FeynCalc, Package X, DRalgo, PTPlot, PT2GWFinder, MESA Stellar Astrophysics, Linux/Unix